Applied Economist and Data Scientist

Building the Data Policy Questions Need

Rodrigo França Chaves

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Website · Google Scholar · LinkedIn · GitHub · Brazilian citizen, F1 visa (STEM OPT eligible)

Applied economist and data scientist working on regulated sectors: who owns water, power and housing, how they are taxed and governed, and what that does to the households on the other end. Most of my work starts with building data that did not exist in usable form, including police district boundaries reconstructed from geocoded incidents, century-old insurance atlases read with a vision-language model, satellite grids over a new transit corridor, and 1.1 million property tax records joined across six fiscal years. First author of a peer-reviewed article in *Utilities Policy* on the health effects of water privatization in Brazil, where the results varied across municipalities and we reported them that way. I teach the graduate machine learning course at Harris.

EDUCATION

M.S. in Computational Analysis and Public Policy *Sep 2025 – Jun 2027*

The University of Chicago, Chicago, IL

Specializations: Markets & Regulation; Health Policy

Completed coursework: Economic Development and Policy; Machine Learning for Public Policy; Advanced Statistics for Data Analysis I and II; Databases for Public Policy; Computer Science with Applications I and II

Postgraduate Degree in Data Science (Pós-Graduação) *2024*

Insper – Institute of Research and Education, São Paulo, Brazil

Coursework: Statistical Inference, Econometrics, Machine Learning, Cloud Computing, Predictive Modeling

B.A. in International Relations *2022*

Ibmec – Brazilian Institute of Capital Markets, Belo Horizonte, Brazil

Coursework: Political Economy, International Trade, Macroeconomics, Quantitative Methods, Public Policy, Political Science

PUBLICATIONS & WORKING PAPERS

Private Ownership of Water and Wastewater Systems: Assessing Health Impacts

Utilities Policy, peer-reviewed journal article (first author)

Brazil's 2020 sanitation law made it substantially easier for private operators to run water and wastewater systems. This paper asks whether the earlier wave of privatizations justified that decision, comparing propensity-matched municipalities before and after transfer between 1998 and 2021 with Callaway and Sant'Anna staggered difference-in-differences. We find significant reductions in hospital admissions for diarrheal disease among children under five (ATT -11.6 for infants, -15.8 at ages 1 to 5) and no effect on diseases unrelated to sanitation. Results vary across municipalities; the most robust evidence comes from Tocantins, the only state to move to full private ownership. I wrote the paper, assembled the data, and wrote the analysis code; my co-author supervised.

Body-Worn Cameras and Property Crime: Reduced-Form Evidence from São Paulo

Under review (second and corresponding author)

Between 2021 and 2022 São Paulo deployed 10,125 body cameras across 64 police battalions. Using confidential Military Police records from 2020 to 2023, we estimate what changed. Battalion boundaries are not published, so I reconstructed them from geocoded incidents, denoised and dissolved into Voronoi polygons, which made spatial treatment assignment possible. Vehicle theft rises after adoption; confrontations fall but lose significance once we condition on the baseline crime that drove the rollout. We report the theft result as a robust reduced-form association rather than a causal effect, since the pre-test rejects and a sensitivity analysis breaks down below pre-treatment violations. I did the methodology, data, code and analysis.

WORK EXPERIENCE

Teaching Assistant, Machine Learning for Public Policy *Sep 2026 – Present*

Harris School, University of Chicago, Chicago, IL

- PPHA 30545, the graduate machine learning course: sets the weekly quizzes, runs the lab sessions, and works through the previous week's answers with students, plus office hours for one-to-one help. Took the course in Spring 2026 and joined the teaching team the next quarter.

Graduate Research Assistant *Mar – Jun 2026 · Sep 2026 – Present*

Mansueto Institute for Urban Innovation, UChicago, Chicago, IL

- Co-author on a methods and data paper in preparation, turning the scanned index pages of century-old Sanborn fire insurance atlases into structured records. Pages are read with a vision-language model rather than conventional OCR, through a provider-agnostic runner. Built the evaluation behind it: hand-labeled ground truth, blind validation windows, and measured precision and recall across competing prompts, which is what established the production

prompt over the alternatives.
- Linked the extracted sites to the full-count US census, 1900 to 1940, by block, producing a record of industrial location in twentieth-century Chicago that did not previously exist.

Data Science and Research Intern, DECDI (formerly DIME) *Jun – Aug 2026*
World Bank, Washington, DC

- Measured how construction in Dakar responded to a new bus rapid transit corridor, using difference-in-differences across 100 m grid cells built from Google Open Buildings and GHSL satellite data. Because the corridor was announced in 2018 and opened in 2024, the design identifies the anticipation and construction-phase response rather than the effect of operation.
- Identified that the project's building-detection threshold had been set at nearly twice the value its source paper recommends for Africa. The setting discarded real structures most heavily in dense and informal areas, a bias that varies by neighborhood and so does not difference out. Shipped the corrected pipeline with 200 tests and an interactive explorer.
- Reconstructed four decades of Mumbai property tax and land-use policy into a single timeline, then tested millions of assessment records for bunching at policy thresholds, the signature of manipulated declared values.
- Quantified how Mumbai's market for additional floor area divides across regulatory instruments (TDR, Premium FSI, Fungible FSI) over two decades of transactions, and which market, regulatory, and political conditions drive that choice. *(Findings confidential to the World Bank.)*

Research Assistant & Teaching Assistant *Aug 2023 – Jul 2025*
Insper (Cities Lab, Arq.Futuro), São Paulo, Brazil

- Two years of quasi-experimental policy evaluation across water utilities, crime, property taxation, and real estate markets, using difference-in-differences, regression discontinuity, and spatial methods.
- Constructed the underlying data: home prices, tax assessments, and disease rates matched across municipalities and census tracts, reconciling administrative records from several government sources so the panels aligned across two decades. Without that step the privatization analysis would not have been identifiable.
- First author on the resulting *Utilities Policy* article and second author on a working paper on police body cameras.
- Held concurrently with the research role: taught GIS and R to more than 50 students, advised undergraduate theses through Brazil's required capstone, one of which placed second in its cohort, and ran an intensive field course with municipal officials.

PROJECTS

NYC Property Tax: A Horizontal-Equity Audit `Python · LightGBM · scikit-learn`
Horizontal equity requires that similar properties carry similar tax burdens. This project tests whether New York's assessment roll meets that standard across 1,097,559 properties. A citywide benchmark is uninformative when a Manhattan co-op and a Queens two-family fall into the same comparison, so each property is measured against a peer group defined on six dimensions with a five-level fallback. On that definition roughly 43% sit more than 15% from their peer-group median; both the peer definition and the threshold are ours, so the figure measures dispersion within comparable groups rather than proven misassessment. Narrowing the value bands from quintiles to ventiles cut within-group price spread from $475k to $48k for a small accuracy loss. Built on 138 leakage-free features and LightGBM, delivered as a dashboard for audit triage. On a three-person course project I designed the peer-group methodology, built the pipeline, and trained the models. [GitHub] [Live Dashboard]

Data Centers Next Door: Cloud Infrastructure and Housing Costs `R · Shiny · Spatial Econometrics`
Data center construction around Chicago has accelerated, raising the question of what it does to nearby housing and household costs. I scraped 124 listed facilities, confirmed and geocoded 45, and dated each opening from air permit filings between 2000 and 2024. Those dates are matched against ZIP-level Zillow home values and Census household and utility costs across 660 metropolitan ZIP codes, so a user can examine what moved around each opening. The design is descriptive rather than causal: a facility opening near rising prices is not evidence that it caused them. On a four-person course project I did the scraping, geocoding, and Shiny dashboard. [GitHub] [Live Dashboard]

TECHNICAL SKILLS

Methods: Causal Inference (DiD, Staggered DiD, RDD, IV), Panel & Spatial Econometrics, Remote Sensing
Programming & Data: Python, R, SQL, PostgreSQL, Stata; data pipelines, OCR & vision-language extraction, geospatial workflows
Tools: QGIS, ArcGIS, Git, LaTeX, Excel, Streamlit, Shiny, DuckDB
Domain: Property tax & valuation, land use, housing & infrastructure policy, utility regulation
Languages: Portuguese (Native), English (Fluent, TOEFL 112), Spanish (Professional), German (Conversational)